

hustreport.cls

HUST University Report Template Class

Complete Source Code Analysis Tutorial

Version 3.0

2026 年 2 月 3 日

This document provides a comprehensive module-by-module analysis of the hustreport.cls class file.

目录

第一章	Introduction and Overview	4
1.1	Template Purpose	4
1.2	Architecture Overview	4
1.3	Compilation Requirements	4
第二章	Module 0-1: Version Declaration and Class Options	5
2.1	Version Declaration (Lines 1-14)	5
2.2	Boolean Switches (Lines 19-24)	5
2.3	Option Declaration (Lines 27-31)	5
2.4	Option Forwarding and Processing (Lines 33-35)	6
2.5	Base Class Loading (Lines 37-43)	6
第三章	Module 2: Package Loading	7
3.1	Organization Philosophy	7
3.2	Page and Layout Packages (Lines 51-55)	7
3.3	Title and TOC Packages (Lines 57-59)	7
3.4	Mathematics and Physics (Lines 61-66)	7
3.5	Graphics and Color (Lines 68-72)	7
3.6	Tables and Lists (Lines 74-84)	7
第四章	Module 3: Information Interface	9
4.1	Internal Storage Variables (Lines 97-102)	9
4.2	Public Interface Commands (Lines 105-110)	9
4.3	Task Book Variables (Lines 113-116)	9
4.4	Smart Underline Component (Lines 119-123)	10

第五章	Module 4: Smart Font Configuration	11
5.1	Design Philosophy	11
5.2	Font Reset (Lines 129-131)	11
5.3	English Font Configuration (Lines 147-167)	11
5.4	Chinese Font Configuration (Lines 171-254)	12
第六章	Module 5-6: Page Layout and Section Titles.....	13
6.1	Conditional Page Layout (Lines 260-301)	13
6.2	Counter Configuration (Lines 305-313)	13
6.3	Section Title Styles (Lines 325-352)	14
6.4	TOC Styles (Lines 358-376)	14
第七章	Module 7-8: Figures, Tables, and PDF Properties	16
7.1	Float Parameters Optimization (Lines 386-392)	16
7.2	Caption Styling (Lines 395-410)	16
7.3	Numbering Formats (Lines 415-423)	16
7.4	PDF Properties (Lines 442-448)	17
第八章	Module 9-10: Special Environments and Cover	18
8.1	Chinese Abstract Environment (Lines 455-468)	18
8.2	Bibliography Automation (Lines 471-498)	18
8.3	Appendix Environment (Lines 501-527)	18
8.4	Cover Generation with TikZ (Lines 641-680)	19
第九章	Module 11-12: Advanced Layouts and Code Environments	
20		
9.1	Text-Figure Mixed Layout (Lines 693-737)	20
9.2	Parallel Figures Environment (Lines 739-761)	20
9.3	New Multi-Figure Environments (v3.0)	21
9.3.1	figurerow - Auto-width Row Layout	21
9.3.2	figuregrid - Grid Layout	21
9.4	Code Environment Design (Lines 772-811)	22
9.5	tcolorbox Integration (Lines 814-855)	22

第十章	Module 13-16: Abstract, Algorithm, and Notice Environments	24
10.1	English Abstract (Lines 872-887)	24
10.2	Algorithm Localization (Lines 892-904)	24
10.3	Notice Boxes (Lines 928-957)	24
10.4	Acknowledgement Environment (Lines 962-970)	25
第十一章	Module 17: Professional List Customization	26
11.1	Custom TikZ Bullet Markers (Lines 981-1005)	26
11.2	Itemize Configuration (Lines 1018-1031)	26
11.3	Custom Enumerate Styles (Lines 1045-1070)	27
第十二章	Module 18-19: Utilities and Review System	28
12.1	Directory Tree Display (Lines 1077-1078)	28
12.2	Todo/Fixme/Note Commands (Lines 1100-1139)	28
12.3	PDF Metadata Injection (Lines 1142-1149)	28
12.4	Review System (Lines 1166-1284)	28
第十三章	Module 20: Tutorial Documentation Environments	30
13.1	democode: Side-by-Side Display (Lines 1327-1351)	30
13.2	demovert: Vertical Display (Lines 1356-1377)	30
13.3	API Reference Card (Lines 1400-1418)	30
13.4	Inline Code Helpers (Lines 1438-1440)	31
Summary		32

第一章 Introduction and Overview

1.1 Template Purpose

The `hustreport.cls` is a $\text{\LaTeX}$ document class designed for Heilongjiang University of Science and Technology (HUST) undergraduate students. It provides a complete solution for:

- Social investigation reports
- Internship reports
- Graduation theses (with `thesis` option)

1.2 Architecture Overview

The class file consists of **20 functional modules** spanning approximately 1445 lines of code:

1. **Module 0-1:** Version declaration and class options
2. **Module 2:** Package loading (categorized by function)
3. **Module 3:** User information interface
4. **Module 4:** Smart font configuration
5. **Module 5-6:** Page layout and section titles
6. **Module 7-8:** Figure/table formatting and PDF properties
7. **Module 9-10:** Special environments and cover generation
8. **Module 11-20:** Advanced features (code blocks, review system, etc.)

1.3 Compilation Requirements

This template **must** be compiled with XeLaTeX:

```
xelatex your_document.tex
bibtex your_document      % If using bibliography
xelatex your_document.tex
xelatex your_document.tex
```

第二章 Module 0-1: Version Declaration and Class Options

2.1 Version Declaration (Lines 1-14)

Every $\text{\LaTeX}$ class file must declare its format requirements and provide identification:

```
% Line 12-13: Format and class identification
\NeedsTeXFormat{LaTeX2e}
\ProvidesClass{hustreport}[2026/01/05 v3.0 HUST Report Class]
```

Key Points:

- `\NeedsTeXFormat{LaTeX2e}` ensures compatibility with modern $\text{\LaTeX}$
- `\ProvidesClass` provides version information displayed in logs

2.2 Boolean Switches (Lines 19-24)

The class uses boolean switches to control conditional compilation:

```
% Platform detection switches
\newif\ifhust@oswin    \hust@oswinfalse
\newif\ifhust@osmac    \hust@osmacfalse
\newif\ifhust@oslinux  \hust@oslinuxfalse
\newif\ifhust@optfonts \hust@optfontsfalse
\newif\ifhust@thesis   \hust@thesisfalse
```

Implementation Pattern:

- `\newif` creates a boolean toggle with default false
- Naming convention: `\ifhust@<name>` uses internal prefix `hust@`
- All switches initialize to false, enabling opt-in behavior

2.3 Option Declaration (Lines 27-31)

Options are declared using `\DeclareOption`:

```
% Platform options
\DeclareOption{win}{\hust@oswintrue}
\DeclareOption{mac}{\hust@osmactrue}
\DeclareOption{linux}{\hust@oslinuxtrue}
\DeclareOption{fonts}{\hust@optfontstrue}
\DeclareOption{thesis}{\hust@thesistrue}
```

2.4 Option Forwarding and Processing (Lines 33-35)

Unrecognized options are forwarded to the base class:

```
\DeclareOption*{\PassOptionsToClass{\CurrentOption}{ctexrep}}
\ProcessOptions\relax
```

2.5 Base Class Loading (Lines 37-43)

The template is built on ctexrep for Chinese support:

```
\LoadClass[a4paper,UTF8,zihao=-4,linespread=1.5]{ctexrep}
```

Parameters Explained:

a4paper — Standard A4 paper size (210mm × 297mm)

UTF8 — UTF-8 encoding for Chinese characters

zihao=-4 — Default font size: xiaosi (12pt equivalent)

linespread=1.5 — 1.5x line spacing for readability

第三章 Module 2: Package Loading

3.1 Organization Philosophy

Packages are organized by functional category for maintainability:

3.2 Page and Layout Packages (Lines 51-55)

```
\RequirePackage{geometry} % Page margins
\RequirePackage{fancyhdr} % Headers and footers
\RequirePackage{float} % Float placement control
\RequirePackage[sort&compress,numbers,square]{natbib}
```

3.3 Title and TOC Packages (Lines 57-59)

```
\RequirePackage{titletoc} % TOC formatting
\RequirePackage{titlesec} % Section title formatting
```

3.4 Mathematics and Physics (Lines 61-66)

```
\RequirePackage{amsmath,amssymb,amsfonts,amsthm}
\RequirePackage{physics} % Physics notation shortcuts
\RequirePackage{siunitx} % SI units
\RequirePackage{newtxmath} % Math font optimization
\RequirePackage{calc} % Length calculations
```

3.5 Graphics and Color (Lines 68-72)

```
\RequirePackage{graphicx} % Image inclusion
\RequirePackage{subcaption} % Subfigures
\RequirePackage{tikz} % Drawing and positioning
\RequirePackage[dvipsnames,svgnames,x11names]{xcolor}
```

3.6 Tables and Lists (Lines 74-84)

```
\RequirePackage{booktabs} % Professional tables
```

```

\RequirePackage{threeparttable}
\RequirePackage{caption}
\RequirePackage{tabularx}    % Auto-width tables
\RequirePackage{xltabular}   % Long tables with page breaks
\RequirePackage{multirow}    % Multi-row cells
\RequirePackage{makecell}    % Cell line breaks
\RequirePackage{array}       % Column format extensions
\RequirePackage{enumitem}    % List customization
\RequirePackage{diagbox}     % Diagonal box headers

```

第四章 Module 3: Information Interface

4.1 Internal Storage Variables (Lines 97-102)

The class defines internal variables using @ prefix:

```
\makeatletter
\newcommand{\@major}{} % Major/Program
\newcommand{\@dept}{} % Department
\newcommand{\@advisor}{} % Advisor
\newcommand{\@team}{} % Team members
\newcommand{\@studentID}{} % Student ID
\newcommand{\@reportID}{} % Report number
```

4.2 Public Interface Commands (Lines 105-110)

User-facing commands that set the internal variables:

```
\newcommand{\major}[1]{\def\@major{#1}}
\newcommand{\dept}[1]{\def\@dept{#1}}
\newcommand{\advisor}[1]{\def\@advisor{#1}}
\newcommand{\team}[1]{\def\@team{#1}}
\newcommand{\studentID}[1]{\def\@studentID{#1}}
\newcommand{\reportID}[1]{\def\@reportID{#1}}
```

4.3 Task Book Variables (Lines 113-116)

Predefined variables for task book content:

```
\newcommand{\taskTargetText}{Content pending}
\newcommand{\taskContentText}{Content pending}
\newcommand{\taskExpectedText}{Content pending}
\newcommand{\taskScheduleText}{Schedule pending}
```

4.4 Smart Underline Component (Lines 119-123)

Adaptive underline for form fields:

```
\newcommand{\ulineinfo}[2][280pt]{%  
  \begin{minipage}[t]{#1}%  
    \centering\zihao{3}\songti\uline{#2\hfill}%  
  \end{minipage}%  
}
```

第五章 Module 4: Smart Font Configuration

5.1 Design Philosophy

The font system implements a **fallback chain** with three priority levels:

1. **Highest:** Local fonts/ directory
2. **Medium:** Platform-specific system fonts
3. **Fallback:** Fandol open-source fonts

5.2 Font Reset (Lines 129-131)

Clear existing font definitions to prevent conflicts:

```
\let\songti\relax
\let\heiti\relax
\let\kaiti\relax
```

5.3 English Font Configuration (Lines 147-167)

```
\IfFileExists{fonts/times.ttf}{
  % Local Times font found
  \setmainfont{times.ttf}[
    Path = fonts/,
    BoldFont = timesbd.ttf,
    ItalicFont = timesi.ttf,
    BoldItalicFont = timesbi.ttf
  ]
}{
  % System fallback
  \ifhust@oswin
    \setmainfont{Times New Roman}
  \else
    \ifhust@osmac
      \setmainfont{Times New Roman}
```

```

        \else
            \setmainfont{TeX Gyre Termes}
        \fi
    \fi
}

```

5.4 Chinese Font Configuration (Lines 171-254)

The Chinese font setup uses conditional checks for each font file:

```

\IfFileExists{fonts/SimSun.ttf}{
    % Local Chinese fonts available
    \setCJKmainfont{SimSun.ttf}[
        Path = fonts/,
        BoldFont = \hust@CJK@bold,
        ItalicFont = \hust@CJK@italic,
        AutoFakeBold = \hust@CJK@fakebold
    ]
    % ... additional font family definitions
}{
    % Platform-specific system fonts
    \ifhust@oswin
        % Windows: SimSun, SimHei, KaiTi
    \else\ifhust@osmac
        % macOS: Songti SC, Heiti SC, Kaiti SC
    \else
        % Linux: Fandol fonts
    \fi\fi
}

```

第六章 Module 5-6: Page Layout and Section Titles

6.1 Conditional Page Layout (Lines 260-301)

The layout switches based on the thesis option:

```
\ifhust@thesis

% Thesis mode: special header with logo
\geometry{
    a4paper,
    top=3.7cm, bottom=2.5cm,
    left=3.9cm, right=2.5cm,
    headheight=3cm, headsep=0.5cm
}

% Custom header with school logo
\pagestyle{fancy}
\fancyhf{}
\renewcommand{\headrulewidth}{2.5pt}
\fancyhead[L]{\includegraphics[height=2.6cm]{logo.png}}
\fancyhead[C]{...school name and title...}
\fancyfoot[C]{\thepage}
\else
% Standard report mode
\geometry{left=3.017cm, right=3.017cm,
    top=2.54cm, bottom=2.54cm}
\pagestyle{plain}
\fi
```

6.2 Counter Configuration (Lines 305-313)

Section numbering uses Arabic numerals with chapter binding:

```
\renewcommand{\thesection}{\arabic{section}}
\renewcommand{\thesubsection}{\thesection.\arabic{subsection}}
```

```
\renewcommand{\thesubsubsection}{\thesubsection.\arabic{subsubsection}}
```

```
% Bind counters to sections (reset per chapter)
```

```
\counterwithin{equation}{section}
```

```
\counterwithin{figure}{section}
```

```
\counterwithin{table}{section}
```

6.3 Section Title Styles (Lines 325-352)

Using ctexset for Chinese-aware formatting:

```
\ctexset{
  section = {
    number = \chinese{section},
    name = {Chapter },
    format = \centering\zihao{3}\heiti\bfseries,
    aftername = \quad,
    beforekip = 15pt, afterskip = 20pt
  },
  subsection = {
    format = \zihao{4}\heiti\bfseries,
    aftername = \hspace{1em}
  },
  subsubsection = {
    format = \zihao{-4}\songti,
    aftername = \hspace{1em}
  }
}
```

6.4 TOC Styles (Lines 358-376)

Custom table of contents formatting:

```
\titlecontents{section}[0pt]
```

```
{\addvspace{6pt}\bfseries\heiti\zihao{4}}
```

```
{\thecontentslabel\quad}  
{}  
{\titlerule*[0.5pc]{.}\contentspage}
```

第七章 Module 7-8: Figures, Tables, and PDF Properties

7.1 Float Parameters Optimization (Lines 386-392)

Solving the “figure jumping” problem:

```
\renewcommand{\topfraction}{0.85}    % Top floats: 85%
\renewcommand{\bottomfraction}{0.70} % Bottom floats: 70%
\renewcommand{\textfraction}{0.10}   % Min text: 10%
\renewcommand{\floatpagefraction}{0.50}
```

7.2 Caption Styling (Lines 395-410)

Professional caption formatting:

```
\DeclareCaptionFont{heiti}{\heiti}
\DeclareCaptionFont{fivehao}{\zihao{5}}
\captionsetup{
  format=hang,
  labelsep=quad,
  font={fivehao,heiti},
  skip=6pt
}
```

7.3 Numbering Formats (Lines 415-423)

Chapter-relative numbering:

```
\renewcommand{\theequation}{\thesection-\arabic{equation}}
\renewcommand{\thefigure}{\thesection-\arabic{figure}}
\renewcommand{\thetable}{\thesection-\arabic{table}}

% Auto-prefix for references
\labelformat{figure}{Figure~#1}
\labelformat{table}{Table~#1}
\labelformat{equation}{Eq.~(#1)}
```

7.4 PDF Properties (Lines 442-448)

Hyperref configuration:

```
\hypersetup{
  hidelinks,
  colorlinks=false,
  bookmarksnumbered=true,
  bookmarksopen=true,
  pdfstartview=FitH
}
```

第八章 Module 9-10: Special Environments and Cover

8.1 Chinese Abstract Environment (Lines 455-468)

```
\newenvironment{cnabstract}[1]{
  \def\@keyword{#1}
  \clearpage
  \phantomsection
  \addcontentsline{toc}{section}{Abstract}
  \begin{center} \heiti\zihao{3} \@title \end{center}
  \vspace{1ex}
  \begin{center} \heiti\zihao{4} Abstract \end{center}
  \par
}{
  \par\vspace{2ex}
  \noindent{\heiti\zihao{-4} Keywords: }\@keyword
  \newpage
}
```

8.2 Bibliography Automation (Lines 471-498)

```
\newcommand{\MyBibliography}[1]{
  \clearpage
  \phantomsection
  \addcontentsline{toc}{section}{References}
  \renewcommand{\section}[2]{}% Suppress auto-heading
  \bibliographystyle{plain}
  \bibliography{#1}
}
```

8.3 Appendix Environment (Lines 501-527)

```
\newenvironment{appendixblock}{
  \clearpage
```

```

\setcounter{section}{0}

\renewcommand{\thesection}{\Alph{section}}

\ctexset{
  section = {
    name = {Appendix },
    format = \centering\zihao{4}\heiti\bfseries
  }
}
}{}
```

8.4 Cover Generation with TikZ (Lines 641-680)

The cover uses TikZ for absolute positioning:

```

\newcommand{\makecover}{
  \begin{titlepage}
    \begin{tikzpicture}[remember picture, overlay]
      % Logo positioning
      \node[anchor=north east, xshift=-2cm, yshift=-2cm]
        at (current page.north east) {
          \includegraphics[width=0.25\textwidth]{logo.png}
        };
    \end{tikzpicture}
    % Title and info table...
  \end{titlepage}
}
```

第九章 Module 11-12: Advanced Layouts and Code Environments

9.1 Text-Figure Mixed Layout (Lines 693-737)

The `textfigure` environment uses `xparse`:

```
\NewDocumentEnvironment{textfigure}{0{right} m m 0{0.4}}{%  
  \def\tf@pos{#1}%    % Position: left or right  
  \def\tf@path{#2}%    % Image path  
  \def\tf@cap{#3}%    % Caption  
  \setlength{\tf@imagew}{#4\textwidth}%  
  % Calculate text width  
  \setlength{\tf@textw}{\textwidth}%  
  \addtolength{\tf@textw}{-\tf@imagew}%  
  \addtolength{\tf@textw}{-1.5em}%  
  % Begin text capture...  
}{%  
  % Render image + text side by side  
}
```

9.2 Parallel Figures Environment (Lines 739-761)

The `parallelfigures` environment uses `subfigure` for side-by-side images with automatic (a), (b), (c) numbering:

```
\NewDocumentEnvironment{parallelfigures}{m}{%  
  \renewcommand{\pf@maincap}{#1}%  
  \begin{figure}[H]  
    \centering  
    \newcommand{\addfig}[3][0.48]{%  
      \begin{subfigure}[b]{##1\textwidth}  
        \centering  
        \includegraphics[width=\linewidth]{##2}
```

```

        \caption{##3}
    \end{subfigure}%
    \hspace{\fill}
}%
}{%
    \caption{\pf@maincap}%
\end{figure}%
}

```

Subfigure Numbering:

- Main figure: Chapter-relative numbering (e.g., Figure 9-1)
- Subfigures: Automatic (a), (b), (c) format
- Defined in line 415: `\renewcommand\thesubfigure{(\alph{subfigure})}`

9.3 New Multi-Figure Environments (v3.0)

Two new environments for flexible layouts:

9.3.1 `figurerow` - Auto-width Row Layout

```

\begin{figurerow}{Main Title}[columns]
    \figitem{path}{subcaption}
    \figitem{path}{subcaption}
\end{figurerow}

```

9.3.2 `figuregrid` - Grid Layout

```

\begin{figuregrid}{Main Title}
    \gridrow{
        \gridfig[width]{path}{subcaption}
        \gridfig[width]{path}{subcaption}
    }
\end{figuregrid}

```

9.4 Code Environment Design (Lines 772-811)

Color definitions for syntax highlighting:

```
\definecolor{hust@codeback}{RGB}{250,250,250}
\definecolor{hust@keyword}{RGB}{0,92,197}
\definecolor{hust@comment}{RGB}{80,161,79}
\definecolor{hust@string}{RGB}{227,17,108}

\lstdefinestyle{huststyle}{
  basicstyle=\small\ttfamily,
  keywordstyle=\bfseries\color{hust@keyword},
  commentstyle=\itshape\color{hust@comment},
  stringstyle=\color{hust@string},
  numbers=left,
  breaklines=true,
  keepspaces=true
}
```

9.5 tcolorbox Integration (Lines 814-855)

Mac-style code block with traffic lights:

```
\tcbset{
  hustcodestyle/.style={
    enhanced, breakable,
    colback=hust@codeback,
    title={\textbf{Code \thehustcode:}\#1},
    % Mac-style decoration
    overlay unbroken and first={
      \draw[fill=red!70!white]
        ([xshift=6mm,yshift=-1.5mm]frame.north west)
        circle (2.5pt);
      \draw[fill=yellow!70!white]
        ([xshift=11mm,yshift=-1.5mm]frame.north west)
```

```

        circle (2.5pt);
\draw[fill=green!70!white]
    ([xshift=16mm,yshift=-1.5mm]frame.north west)
    circle (2.5pt);
    }
}
}

```

第十章 Module 13-16: Abstract, Algorithm, and Notice Environments

10.1 English Abstract (Lines 872-887)

```
\newenvironment{enabstract}[1]{
  \def\@enkeyword{#1}
  \clearpage
  \addcontentsline{toc}{section}{Abstract}
  \begin{center}\bfseries\zihao{4} Abstract \end{center}
  \linespread{1.5}\selectfont
  \zihao{-4}
}{
  \par\vspace{2ex}
  \noindent\textbf{Keywords: } \@enkeyword
  \newpage
}
```

10.2 Algorithm Localization (Lines 892-904)

```
\RequirePackage{algorithm}
\RequirePackage{algorithmicx}
\RequirePackage{algpseudocode}

\floatname{algorithm}{Algorithm}
\renewcommand{\algorithmicrequire}{\textbf{Input:}}
\renewcommand{\algorithmicensure}{\textbf{Output:}}
\algorenewcommand\algorithmicindent{2.0em}
```

10.3 Notice Boxes (Lines 928-957)

Three themed boxes using tcolorbox:

```
% Notice box (blue theme)
```

```

\newtcolorbox{notice}[1]{
    commonbox,
    colback=Azure1, colframe=DodgerBlue3,
    title={[Notice] #1}
}

% Definition box (green theme)
\newtcolorbox{definition}[1]{
    commonbox,
    colback=Honeydew1, colframe=ForestGreen,
    title={[Definition] #1}
}

% Conclusion box (red theme)
\newtcolorbox{conclusion}[1]{
    commonbox,
    colback=LavenderBlush1, colframe=Crimson,
    title={[Conclusion] #1}
}

```

10.4 Acknowledgement Environment (Lines 962-970)

```

\newenvironment{acknowledgement}{
    \clearpage
    \addcontentsline{toc}{section}{Acknowledgement}
    \section*{Acknowledgement}
    \zihao{-4}\songti\linespread{1.5}\selectfont
}{
    \par
}

```

第十一章 Module 17: Professional List Customization

11.1 Custom TikZ Bullet Markers (Lines 981-1005)

```
% Level 1: Solid square
\newcommand{\hust@item@square}{%
    \tikz[baseline=-0.6ex]
    \node[fill=black, shape=rectangle,
        inner sep=0pt, minimum size=0.5em] {};%
}

% Level 2: Gray circle
\newcommand{\hust@item@circle}{%
    \tikz[baseline=-0.6ex]
    \node[fill=darkgray, shape=circle,
        inner sep=0pt, minimum size=0.35em] {};%
}

% Level 3: Dash line
\newcommand{\hust@item@dash}{%
    \tikz[baseline=-0.6ex]
    \draw[line width=0.8pt] (0,0) -- (0.5em,0);%
}
```

11.2 Itemize Configuration (Lines 1018-1031)

```
\setlist[itemize,1]{
    label=\protect\hust@item@square,
    leftmargin=2.0em, labelsep=0.6em
}

\setlist[itemize,2]{
    label=\protect\hust@item@circle,
    leftmargin=1.8em, labelsep=0.6em
}
```

```

}
\setlist[itemize,3]{
    label=\protect\hust@item@dash,
    leftmargin=1.8em, labelsep=0.6em
}

```

11.3 Custom Enumerate Styles (Lines 1045-1070)

```

% Parenthesized: (1), (2), (3)
\newlist{enumpar}{enumerate}{1}
\setlist[enumpar]{label=(\arabic*)}

% Half-parenthesized: 1), 2), 3)
\newlist{enumbra}{enumerate}{1}
\setlist[enumbra]{label=\arabic*)}

% Circled numbers
\newlist{enumcircled}{enumerate}{1}
\setlist[enumcircled]{
    label=\protect\hust@circled{\arabic*}
}

```

第十二章 Module 18-19: Utilities and Review System

12.1 Directory Tree Display (Lines 1077-1078)

```
\RequirePackage{dirtree}
\renewcommand{\DTstyle}{\ttfamily\small}
```

12.2 Todo/Fixme/Note Commands (Lines 1100-1139)

Breakable note boxes:

```
\newcommand{\todo}[1]{%
  \begin{tcolorbox}[
    hust@todostyle,
    colback=orange!5,
    colframe=orange!80!black,
    title={TODO (Pending)}
  ]
  #1
  \end{tcolorbox}%
}
```

12.3 PDF Metadata Injection (Lines 1142-1149)

```
\AtBeginDocument{
  \hypersetup{
    pdftitle={\@title},
    pdfauthor={\@author},
    pdfsubject={HUST Report/Thesis},
    pdfcreator={HUST Report LaTeX Class}
  }
}
```

12.4 Review System (Lines 1166-1284)

Using pgfkeys for option parsing:

```

\pgfkeys{
  /hustreview/.is family, /hustreview,
  style/.store in = \hust@review@style,
  color/.store in = \hust@review@color,
  side/.store in   = \hust@review@side,
  offset/.store in = \hust@review@offset
}

\NewDocumentCommand{\review}{ O{} m +m }{%
  \pgfkeys{/hustreview, #1}%
  % Render mark based on style (underline/wave/etc.)
  % Draw side note using TikZ overlay
}

```

第十三章 Module 20: Tutorial Documentation Environments

13.1 democode: Side-by-Side Display (Lines 1327-1351)

```
\newtcblisting{democode}[1]{  
    enhanced, breakable,  
    title={#1},  
    listing engine=listings,  
    sidebyside,  
    sidebyside align=top,  
    lefthand ratio=0.5,  
    bicolor,  
    colbacklower=hust@demo@resultbg  
}
```

13.2 demovert: Vertical Display (Lines 1356-1377)

```
\newtcblisting{demovert}[1]{  
    enhanced, breakable,  
    listing engine=listings,  
    listing and text, % Code above, result below  
    bicolor,  
    colbacklower=hust@demo@resultbg  
}
```

13.3 API Reference Card (Lines 1400-1418)

```
\newcommand{\apibox}[3]{%  
    \begin{tcolorbox}[  
        colback=hust@api@bg,  
        colframe=hust@api@frame,  
        title={\textbackslash #1}  
    ]
```

```

\textbf{Parameters:} \texttt{#2} \\
\textbf{Description:} #3
\end{tcolorbox}
}

```

13.4 Inline Code Helpers (Lines 1438-1440)

```

\newcommand{\cmd}[1]{\texttt{\textbackslash #1}}
\newcommand{\env}[1]{\texttt{#1}}
\newcommand{\opt}[1]{\texttt{[#1]}}

```

Summary

This document has provided a comprehensive analysis of the `hustreport.cls` class file, covering all 20 functional modules:

1. Version declaration and class options
2. Package loading organized by function
3. User information interface
4. Smart font configuration with fallback chain
5. Conditional page layout (report vs. thesis)
6. Section title and TOC styling
7. Float optimization and caption formatting
8. PDF properties and hyperlinks
9. Special environments (abstract, bibliography, appendix)
10. Cover and task book generation with TikZ
11. Advanced text-figure layouts
12. Professional code blocks with syntax highlighting
13. English abstract environment
14. Algorithm environment localization
15. Themed notice/definition/conclusion boxes
16. Acknowledgement environment
17. Custom list markers with TikZ
18. Utility features (dirtree, todo, metadata)
19. Advanced review and annotation system
20. Tutorial documentation environments

2026 年 2 月 3 日